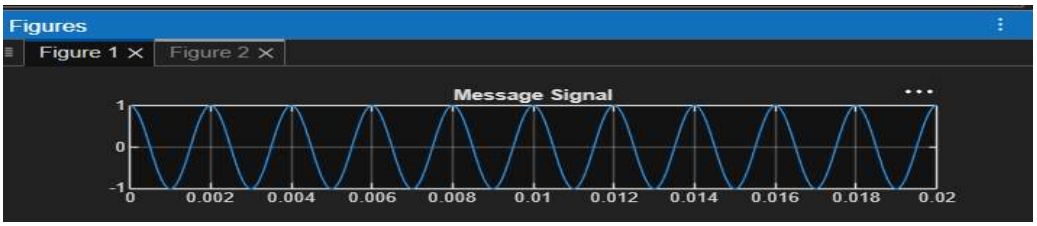
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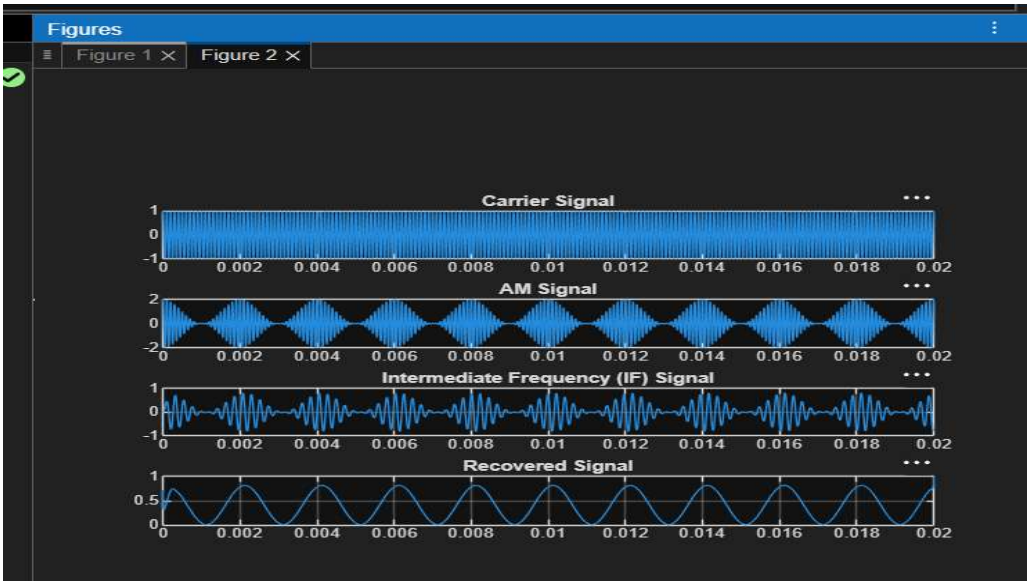
Exp 4: MATLAB SIMULATION OF SUPERHETERODYNE RECEIVER FOR AM COMMUNICATION SYSTEM WITH FREQUENCY MIXING AND FILTERING

Aim:

i) To model and simulate the operation of a superheterodyne receiver for an AM communication system using MATLAB by implementing frequency mixing and intermediate frequency (IF) signal generation.

ii) To recover the original message signal through intermediate frequency filtering and envelope detection, and analyze the waveforms of the message, carrier, AM, IF, and recovered signals.



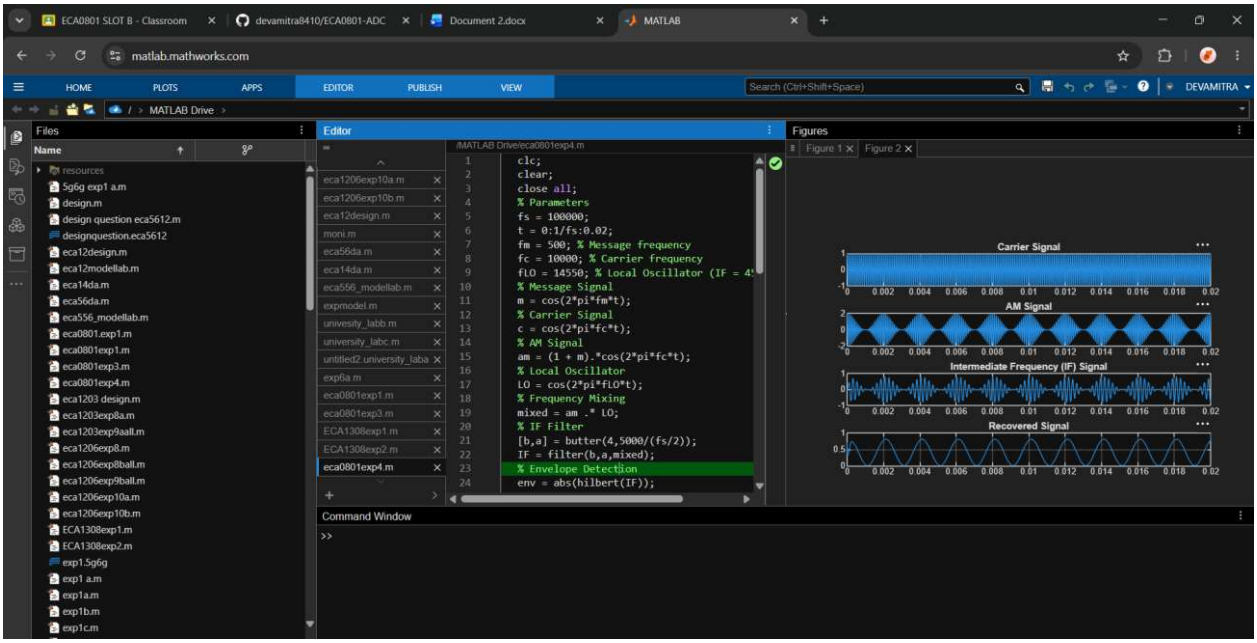


Program:

```
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500; % Message frequency
fc = 10000; % Carrier frequency
fLO = 14550; % Local Oscillator (IF = 4550 Hz)
% Message Signal
m = cos(2*pi*fm*t);
% Carrier Signal
c = cos(2*pi*fc*t);
% AM Signal
am = (1 + m).*cos(2*pi*fc*t);
% Local Oscillator
LO = cos(2*pi*fLO*t);
% Frequency Mixing
mixed = am .* LO;
% IF Filter
[b,a] = butter(4,5000/(fs/2));
IF = filter(b,a,mixed);
% Envelope Detection
env = abs(hilbert(IF));
% Display
figure;
subplot(4,1,1)
plot(t,m)
title('Message Signal')
grid on
figure;
subplot(5,1,2)
plot(t,c)
title('Carrier Signal')
grid on
subplot(5,1,3)
plot(t,am)
title('AM Signal')
grid on
subplot(5,1,4)
plot(t,IF)
title('Intermediate Frequency (IF) Signal')
grid on
subplot(5,1,5)
plot(t,env)
```

```
title('Recovered Signal')
grid on
```

Output:



SAMPLE LAB EXAM QUESTIONS:

1. Using MATLAB simulation, for the given parameters, Carrier Amplitude, V, Carrier Frequency, kHz, Message Amplitude, V, Message Frequency, Hz, Sampling Frequency, kHz, Local Oscillator Frequency, kHz; do the following tasks. Tasks

- Generate the message signal and carrier signal.
- Generate the AM signal using amplitude modulation.
- Generate the local oscillator (LO) signal.
- Perform frequency mixing to obtain the mixed signal.
- Apply a Butterworth filter to obtain the Intermediate Frequency (IF) signal.
- Recover the original message signal using an envelope detector.
- Plot the message signal, carrier signal, AM signal, Intermediate Frequency (IF) signal, and recovered signal, and compare the original and recovered message signals.

Program:

clc; clear; close all;

% Parameters Ac = 5; % Carrier amplitude (V) fc = 100e3; % Carrier frequency (Hz) Am = 1; % Message amplitude (V) fm = 1e3; % Message frequency (Hz) fs = 1000e3; % Sampling frequency (Hz) fLO = 90e3; % Local oscillator frequency (Hz)

t = 0:1/fs:5e-3;

% 1. Message signal m = Am*cos(2*pi*fm*t);

% 2. Carrier signal c = Ac*cos(2*pi*fc*t);

% 3. AM signal AM = Ac*(1 + 0.8*cos(2*pi*fm*t)).*cos(2*pi*fc*t);

% 4. Local oscillator LO = cos(2*pi*fLO*t);

% 5. Frequency mixing mixed = AM.*LO;

% 6. Butterworth bandpass filter fIF = abs(fc-fLO); [b,a] = butter(4, ... [(fIF-2e3)/(fs/2) (fIF+2e3)/(fs/2)], 'bandpass');

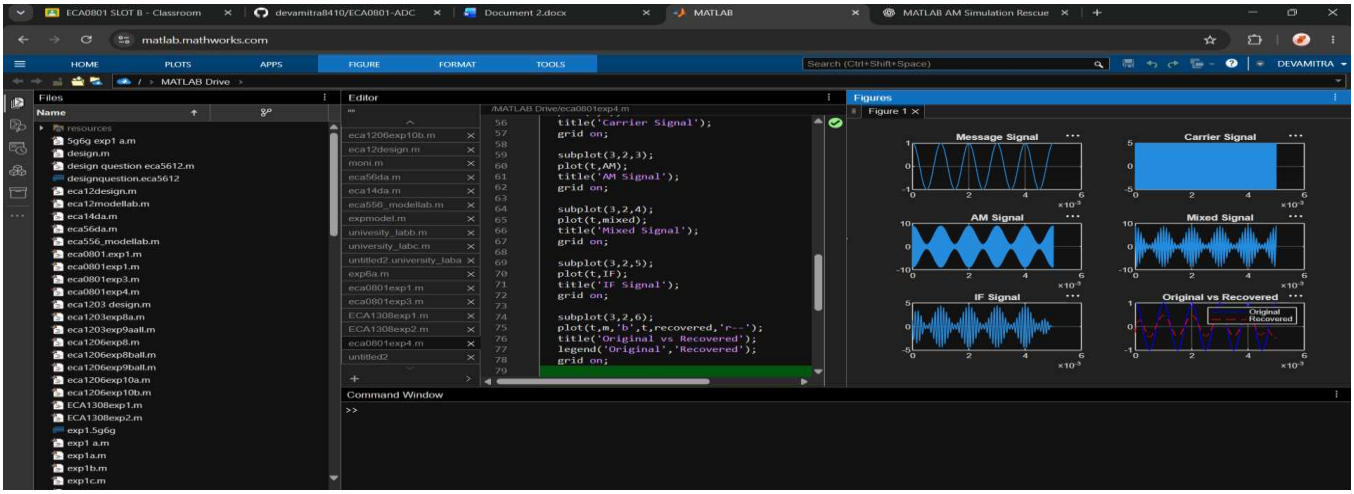
IF = filtfilt(b,a,mixed);

% 7. Envelope detection env = abs(hilbert(IF));

% Remove DC and recover message recovered = env - mean(env); recovered = recovered * ... (max(m)-min(m))/(max(recovered)-min(recovered))/2;

% 8. Plot signals figure;subplot(3,2,1); plot(t,m); title('Message Signal'); grid on;sublot(3,2,2); plot(t,c); title('Carrier Signal'); grid on;subplot(3,2,3); plot(t,AM); title('AM Signal'); grid on;subplot(3,2,4); plot(t,mixed); title('Mixed Signal'); grid on;subplot(3,2,5); plot(t,IF); title('IF Signal'); grid on;subplot(3,2,6); plot(t,m,'b',t,recovered,'r--'); title('Original vs Recovered'); legend('Original','Recovered'); grid on;

Output:



2. Using MATLAB simulation, for the given parameters, Carrier Amplitude, V, Carrier Frequency, kHz, Message Amplitude, V, Message Frequency, kHz, Sampling Frequency, kHz, Local Oscillator Frequency, kHz; do the following tasks. Tasks

- Generate the message signal and carrier signal.
- Generate the AM signal.
- Generate the local oscillator signal and perform frequency mixing.
- Extract the Intermediate Frequency (IF) signal using a Butterworth filter.
- Recover the original message signal using envelope detection.
- Plot the message signal, carrier signal, AM signal, Intermediate Frequency (IF) signal, and recovered signal.
- Compare the original and recovered message signals and comment on the effectiveness of the superheterodyne receiver in recovering the transmitted information.

Program:

```
clc; clear; close all;

% Parameters Ac = 5; % Carrier amplitude (V) fc = 100; % Carrier frequency (kHz) Am = 1; % Message amplitude (V) fm = 1; % Message frequency (kHz) fs = 1000; % Sampling frequency (kHz) fLO = 90; % Local oscillator frequency (kHz)

t = 0:1/fs:5; % Time in ms

% 1. Message and carrier signals m = Amcos(2pifmt); c = Accos(2pifct);

% 2. AM signal AM = Ac*(1 + 0.8cos(2pifmt)).cos(2pifct);

% 3. Local oscillator and frequency mixing LO = cos(2pifLO*t); mixed = AM.*LO;

% 4. Butterworth filter to obtain IF fIF = abs(fc-fLO); [b,a] = butter(4,[(fIF-2)/(fs/2) (fIF+2)/(fs/2)],'bandpass'); IF = filtfilt(b,a,mixed);

% 5. Envelope detection env = abs(hilbert(IF)); recovered = env - mean(env); recovered = recovered * Am/max(abs(recovered));

% 6. Plot signals figure;

subplot(3,2,1); plot(t,m); title('Message Signal'); grid on;

subplot(3,2,2); plot(t,c); title('Carrier Signal'); grid on;

subplot(3,2,3); plot(t,AM); title('AM Signal'); grid on;
```

subplot(3,2,4); plot(t,IF); title('Intermediate Frequency Signal'); grid on;

subplot(3,2,5); plot(t,m,'b',t,recovered,'r--'); title('Original vs Recovered Message'); legend('Original','Recovered'); grid on;

subplot(3,2,6); plot(t,mixed); title('Mixed Signal'); grid on;

% 7. Comparison R = corrcoef(m,recovered); fprintf('Correlation coefficient = %.4f\n',R(1,2));

Output:

